Laboratory Study of Premixed H_2-Air and H_2-N_2-Air Flames in a Low-Swirl Injector for Ultralow Emissions Gas Turbines

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The objective of this study is to conduct laboratory experiments on low-swirl injectors (LSIs) to obtain the basic information for adapting LSI to burn H_2 and diluted H_2 fuels that will be utilized in the gas turbines of the integrated gasification combined cycle coal power plants. The LSI is a novel ultralow emission dry-low NO_x combustion method that has been developed for gas turbines operating on natural gas. It is being developed for fuel-flexible turbines burning a variety of hydrocarbon fuels, biomass gases, and refinery gases. The adaptation of the LSI to accept H_2 flames is guided by an analytical expression derived from the flow field characteristics and the turbulent flame speed correlation. The evaluation of the operating regimes of nine LSI configurations for H_2 shows an optimum swirl number of 0.51, which is slightly lower than the swirl number of 0.54 for the hydrocarbon LSI. Using particle image velocimetry (PIV), the flow fields of 32 premixed H_2-air and H_2-N_2-air flames were measured. The turbulent flame speeds deduced from PIV show a linear correlation with turbulence intensity. The correlation constant for H_2 is 3.1 and is higher than the 2.14 value for hydrocarbons. The analysis of velocity profiles confirms that the near field flow features of the H_2 flames are self-similar. These results demonstrate that the basic LSI mechanism is not affected by the differences in the properties of H_2 and hydrocarbon flames and support the feasibility of the LSI concept for hydrogen fueled gas turbines. [DOI: 10.1115/1.2836480]

Introduction

The motivation of our research is to develop a robust ultralow emission combustor for the gas turbines in FutureGen power plants that burn syngases derived from the gasification of coal. The objective is to adapt a novel low-swirl combustion (LSC) method to these utility size gas turbines that operate on fuels with very high H_2 constituency. LSC is a dry-low-NO_x (DLN) lean premixed combustion method conceived at the Lawrence Berkeley National Laboratory. This technology has been commercialized for industrial heaters by Maxon Corporation of Muncie, IN and is being developed for natural gas and fuel-flexible industrial turbines (5–7 MW) in partnership with Solar Turbines of San Diego, CA. It is also being adapted to 100 kW microturbines in partnership with Elliott Energy Systems of Stuart, FL for integration into a large boiler for combined heat and power generation.

FutureGen power plants utilize the integrated gasification combined cycle (IGCC) approach to produce hydrogen, which is separated from a concentrated CO_2 stream that is then captured for subsequent sequestration. One of its key components is a cost-competitive all-hydrogen fueled turbine with ultralow NO_x emission and high efficiency. To lower NO_x to the FutureGen near zero emission goal of 2 ppm (at 15% O_2), the current approach is to operate the H_2 turbine at lower firing temperatures in combination with selective catalytic reduction. This approach sacrifices efficiency and impacts costs of electricity (via capital cost, efficiency, and capacity output). Therefore, a cost-effective combustion technology that meets the FurtureGen emissions and efficiency targets is critical to achieving its ultimate goal of no more than a 10% increase in the cost of electricity for mature FutureGen-type plants that include CO_2 capture and sequestration.

The purpose of this paper is to conduct laboratory studies of low-swirl injectors (LSIs) burning pure hydrogen and nitrogen-diluted hydrogen fuels to characterize the overall flame behavior, stabilization limits, turbulent flame speeds, and flow field characteristics. The analysis of these data and comparison with those obtained previously for hydrocarbon flames is a necessary first step toward building the scientific and technical foundation for the adaptation of LSI to accept pure and diluted H_2 fuels.

Background

LSC combustion was originally conceived as a research burner for fundamental premixed turbulent flame experiments (e.g., Refs. [1–3]). Through laboratory studies and analyses, its basic operating principle is relatively well understood. LSC supports premixed turbulent flames under a very broad range of operating conditions. Its capability to support robust and stable flames at very fuel lean conditions can be exploited as an ultralow emission combustion technology. LSC operates on the principle of flame propagation. It is therefore not applicable to nonpremixed combustion because the diffusion flames do not propagate.

LSC is an aerodynamic flame stabilization method that utilizes a divergent flow to sustain a freely propagating premixed turbulent flame. The divergent flow is formed only when the swirl intensities are well below the critical vortex breakdown threshold. The linear axial velocity decay region within the divergent flow allows the flame to settle where the local velocity is equal and opposite to the turbulent flame speed. Matching the flow field velocities to the turbulent flame speed is the critical design process. The main advantage of the LSC concept is its very large turndown capacity (demonstrated as 90:1 in laboratory) and its capability to support very lean premixed turbulent flames in a turbulent flow field that allows the flame to withstand swings in operational and mixture conditions without high risk of flame instability, flashback, or blowoff. The flame propagation principle of LSC is fundamentally different from the flame attachment concept

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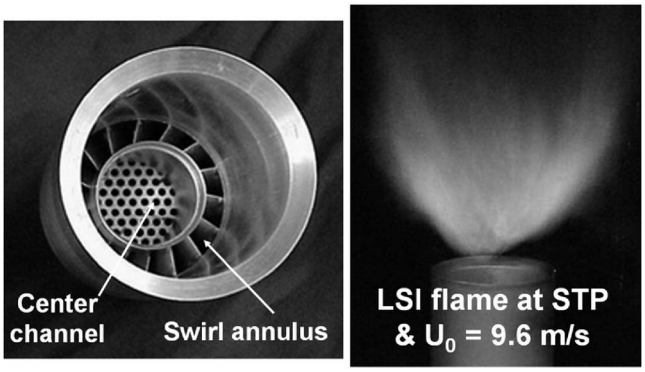

Fig. 1 The first prototype LSI for Taurus 70 engine

of the high-swirl concept utilized in current gas turbine DLN combustors (e.g., GE, Siemens, and Solar) where toroidal vortices with strong recirculation and intense shear turbulence are generated to hold and continuously ignite the turbulent premixed flames.

The key component for a practical implementation of LSC is a patented swirler. This swirler is simpler and easier to operate and control than the first embodiment of LSC using air jets to impart swirl [2,4–6]. Figure 1 shows the first prototype LSI configured for Solar Turbine's 7 MW Taurus 70 (T70) engine [7]. The flow of the premixed reactants through this LSI, as well as all other LSC devices, is separated into two passages: An outer annular with swirl vanes and an open center channel that allows a portion of the reactants to remain unswirled. The unswirled reactants flowing through the center channel is a critical requirement of LSC whose presence retards the recirculation and promotes the formation of flow divergence. The flame is detached from the LSI. Its liftoff position is controlled by the divergence rate that is adjustable by the flow split between the swirled and the unswirled flows. The most convenient means to vary this flow split is by changing the blockage ratio of the perforated screen covering the center channel.

The LSI prototype utilizes a swirler engineered for the T70 $SoLoNO_x$ engines. To configure this swirler into a LSI, the center body was removed and replaced by a perforated plate selected according to the engineering rules developed for industrial low-swirl burners (LSBs). The rules are specified in terms of acceptable ranges of swirl number ($0.4<S<0.55$) and swirler recess ($2<L_i/R_i<3$). The swirl number is defined as

$$S=\frac{2}{3}\tan\alpha\frac{1-R^3}{1-R^2+[m^2(1/R^2-1)^2]R^2} \tag{1}$$

where α is the vane angle, $R=R_c/R_i$, and $m=m_c/m_s$ (where m_c and m_s are, respectively, the mass fluxes of the unswirled and the swirled flows). The $SoLoNO_x$ swirler has an internal radius of 3.175 cm with 16 curved vanes with a discharge angle $\alpha=40$ deg. The ratio of the center channel to injector radii R is 0.63. To meet the design criteria, a 58% blockage perforated plate with circular holes in a hexagonal arrangement was chosen. This LSI has a swirl number $S=0.54$ and generates a flame that has an optimum liftoff height, x_f, of approximately $0.5R_i$ from the nozzle exit. Another key parameter is the swirler recess distance L_i. It controls the residence time for the interaction between the swirled and unswirled flows and has a direct effect on the LSI performance. As in the industrial burners, L_i was set at $L_i/R_i=3$.

Rig tests of the first LSI at elevated temperatures and pressures [7] showed that the LSI flame remained stable and oscillation free throughout the test matrix covering full load (800°F, 15 atm) and partial load (450°F, 6 atm) conditions. The NO_x emissions are 60% lower than those of the $SoLoNO_x$ high-swirl injector. The results demonstrated that the LSI can attain ultralow emissions at conditions farther away from the lean-blowoff (LBO) limit to avoid inciting combustion oscillations. The lowest levels of NO_x (<2 ppm) and CO (<10 ppm) at a firing temperature of 2670°F are comparable with those from much more costly and less durable catalytic options. The LSI achieves the performance targets with only a few design iterations because the knowledge gained from the laboratory experiments is directly useful to guide the turbine hardware design. As the bulk of the optimization can be accomplished in the laboratory, this brings about significant cost savings in the development process.

A fully functional engine-ready LSI prototype has also been developed. It employs a simple fuel spoke injection scheme similar to the one for the current $SoLoNO_x$ injector. For load following, a pilot is imbedded into the center plate. Again, laboratory experiments played a key role in optimizing the fuel injector placement position and the size of the pilot. Single injector tests at simulated gas turbine conditions and multiinjector tests at atmospheric conditions in the T70 annulus combustor liner show that the performance of the engine-ready LSI prototype is similar to the first prototype [8]. A set of the engine-ready LSI prototypes was fabricated and tested successfully in a T70 in June 2006.

The scientific foundation obtained for LSC plays a critical role for its adaptation to gas turbines. In a continuing effort to expand the knowledge base, a recent study focused on gaining better understanding of how the LSI flow field structures and flame properties evolve with Reynolds numbers [9]. These experiments were performed in open flames to circumvent the difficulties and uncertainties associated with laser studies of flames within a quartz enclosure. The conveniences offered by open flame studies facilitate the collection of a large amount of data to examine the overall flame behaviors and flow field evolution that is of significant value to gas turbine development. Moreover, previous studies (e.g., Ref. [10]) showed that the enclosure effects are minimal when the enclosure radius is about three times larger than the injector (or burner) radius. The 3:1 enclosure ratio has been adopted as the guideline for sizing the enclosures for industrial burners and for optimizing the LSI radius for the T70 combustor. Field experiences and rig-test results confirmed that the knowledge gained open flames are directly relevant to the enclosed flames in gas turbines.

Analyses of the velocity measurements obtained from particle image velocimetry (PIV) of these open CH_4 flames show that the flow field generated by the LSI exhibits a self-similarity behavior. Two parameters derived from the centerline velocity profiles of the LSI flow fields are invoked to characterize self-similarity. They are the virtual origin of the divergent flow x_0 and the nondimensional axial aerodynamic stretch rate a_x. To characterize the flame, the turbulent flame speeds S_T and the position of the leading edge of the flame brush, x_f, are used. These parameters can be deduced by using the velocity profiles obtained along the centerline (see Fig. 9 of Ref. [9]).

The analysis of S_T shows a linear dependency on turbulence intensity u'. An analytical equation (Eq. (2)) for the velocity balance at x_f shows that a coupling of the self-similar flow field and linear turbulent flame speed correlations with u' is the reason why the LSI flame remains stationary through a wide range of velocities and fuel air equivalence ratios ϕ.

$$1-\frac{dU}{dx}\frac{(x_f-x_0)}{U_0}=\frac{S_T}{U_0}=\frac{S_L}{U_0}+\frac{Ku'}{U_0} \tag{2}$$

The two terms on the far right hand side simply state that S_T for hydrocarbon flames increases linearly with u' above the base line value of the laminar flame speed S_L at a slope of K. This slope is an empirical correlation constant derived from the experimental measurements[1] and is 2.16 for hydrocarbon flames. The first term

[1]The linear dependency of S_T on u' is not universal as S_T in other burners tends to be nonlinear and shows "bending."

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on the right hand side tends to a small value at large bulk flow velocity U_0 because the laminar flame speeds of typical hydrocarbon fuels are on the order of 0.2–0.7 m/s. The second term on the right hand side is a constant because turbulence in the central region of the LSI is controlled by the perforated plate and scales with U_0. On the left hand side, self-similar means that $a_x = dU/dx/U_0$ in the second term is constant. The main consequence is that the flame position x_f-x_0 has an asymptotic value at large U_0. Therefore, when S_L is held constant at a given ϕ, flame shift, i.e., changes in x_f, occurs only at low velocities where U_0 is in the same order as S_L. When $U_0 \gg S_L$, changing stoichiometry and/or U_0 do not result in a significant flame shift. These flame behaviors have been confirmed by laboratory experiments and testing of industrial burners and gas turbine injectors.

Equation (2) provides an analytical expression that describes the relationship between the LSI flow field and the turbulent flame properties. Because the aerodynamic stretch rate is proportional to the swirl number, this expression gives a first order estimate on how the LSI can be adjusted to accept different fuels. For example, it shows that a significant adjustment may not be necessary for burning different hydrocarbon fuels. This is due to the fact that the S_L and heat release rates are close to those of natural gas. As long as the turbulent flame speeds have a linear correlation with u' at a rate of increase close to that of natural gas flames, the axial aerodynamic stretch rate, i.e., swirler design, does not need to be adjusted to accommodate these other hydrocarbon fuels. Recent laboratory studies of CH_4, C_3H_8, C_2H_4, and CH_4/H_2 flames [9,11] have produced the data to support this conjecture.

To develop the LSI for H_2 fuels, the key question concerns the correlation of the turbulent flame speeds for these flames with significantly higher S_L and preferential diffusion than those of hydrocarbons. Determining the correlation of the S_T for H_2 flames with u' is therefore the critical information needed to adapt LSI for burning H_2. Additionally, characterizing the effects of the H_2 flames on the flow field, particularly how they alter the axial aerodynamic stretch rate in the near field region, is important in gaining insight on predicting the H_2 flame behavior at higher velocities, temperatures, and pressures. The purpose of this investigation is to obtain the experimental data on H_2 flames to begin building a body of knowledge needed for guiding the development of LSI for FutureGen turbines.

Approach

The LSI configuration for the T70 engine is a logical starting point for the evaluation of LSC operation with H_2. Following the same approach for natural gas development, the first step is to explore the operating regime and characterize the performance through laboratory open H_2 flame experiments. The operating regime is defined by the flame stability and LBO limits as a function of U_0. The performance criteria are flame uniformity, flame position, and any other flame behaviors that may affect the operation of the LSI. The discussion on Eq. (2) shows that the LSI can be adapted to accommodate differences in flame properties by adjusting the axial aerodynamic stretch rate. The most convenient way is by changing the blockage ratio of the center-channel plate to vary its swirl number S (i.e., varying m in Eq. (1)). The optimum LSI configuration for H_2 fuels can be then determined by comparing the operating regimes and the performance criteria for a range of S.

Next, the flow field and flame properties of these LSIs are characterized by PIV measurements. The velocity results can be analyzed the same way as those from the hydrocarbon flames to obtain a set of empirical constants for H_2 fuels. In addition, the swirl number effects and the flow field responses to H_2 flames can also be examined.

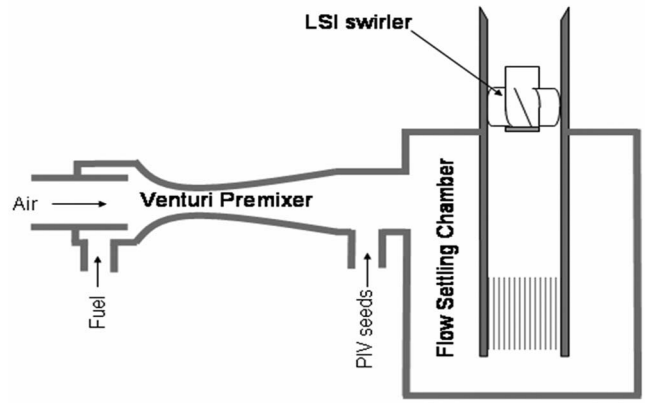

Fig. 2 Schematics of the LSI setup

Experimental Apparatus and Diagnostics

The physical dimensions of the LSI are given above. For the LBO and PIV investigations, the LSIs were mounted vertically on top of a cylindrical settling chamber (Fig. 2). Air supplied by a fan blower enters at the side of a 25.4 cm diameter chamber and flows into the LSI via a centrally placed 30 cm long straight tube. To produce a uniform flow into the LSI, a perforated screen is placed 5 cm upstream. Air flow from a blower is adjusted by a PC controlled valve and monitored by a turbine meter. Fuel is injected in a commercial venturi premixer to ensure a homogeneous mixture for the injector. Both the fuel and the PIV seeder flows are controlled by electronic mass flow controllers and set according to a predetermined value of mixture compositions and ϕ. The fuel and air supply system for this study has a larger capacity than the one used previously [7,9,11] to enable experiments up to $U_0 = 22$ m/s (65 g/s air). Increasing the bulk flow velocity is necessary for the H_2-air flame studies as these flames flash back at a high U_0 than their hydrocarbon counterparts [9]. The experiments utilize research grade H_2 with and without N_2 dilution. Because premixed H_2 flames are invisible to the naked eye, a webcam is used as a safety monitor and to examine the overall flame features. To detect the near-infrared emissions from the H_2 flame, the IR filter of the webcam was removed. The PIV experimental conditions are summarized in Table 1 with the combustion properties calculated using an algorithm by Zhang et al. [12]. Due to the very low adiabatic temperatures of all these H_2 flames, their NO_x emissions are expected to be well below detectable limits and therefore not sampled.

Flow field information for the nonreacting flows and flames was obtained using PIV. To facilitate data collection, the flows and the flames were nonenclosed. The PIV system consists of a New Wave Solo PIV laser with double 120 mJ pulses at 532 nm and a Kodak/Red Lake ES 4.0 digital camera with a 2048×2048 pixel resolution. The optics were configured to capture a field of view of approximately 13×13 cm^2 covering the near field as well as the far field of the flames with a 0.065 mm/pixel resolution. A

Table 1 Mixtures for PIV measurements

Fuel	ϕ	T_{ad} (K)	S_L (m/s)
H_2	0.3	1189	0.12
H_2	0.35	1312	0.18
H_2	0.38	1383	0.22
0.75 H_2/0.25 N_2	0.31	1182	0.09
0.75 H_2/0.25 N_2	0.37	1317	0.14

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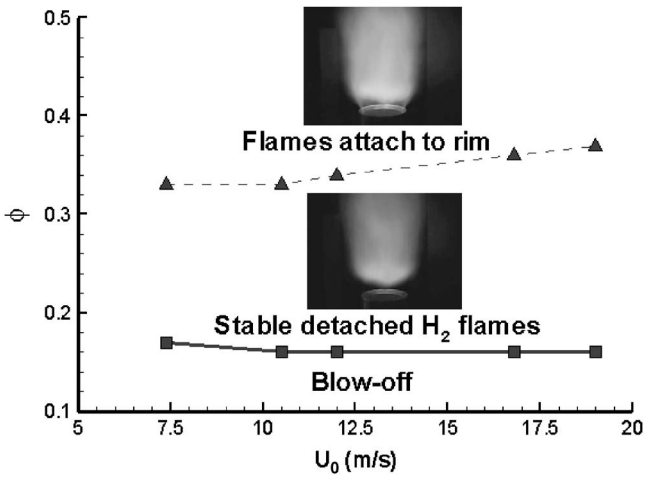

Fig. 3 Operating regime of LSI-R for H_2

cyclone-type particle seeder seeds the air flow with 0.6–0.8 μm Al_2O_3 particles, which should track velocity fluctuations up to 10 kHz [13].

Data acquisition and analysis were performed using a software developed by Wernet [14]. Because of the complex and 3D nature of the swirling flow field, care had to be taken to optimize interframe timing, camera aperture setting, light-sheet thickness, and seed density to ensure high data fidelity. Using a portion of the light sheet with approximately 1.1 mm thickness (away from the 0.3 mm waist produced by the 450 mm spherical lens) and a short interframe time (30 μs) helped to freeze the out-of-plane motion of seed particles. Sets of 224 image pairs were recorded for each experiment corresponding to a minimum criterion required to produce stable mean and rms velocities. The PIV data were processed using 64×64 pixel cross-correlation interrogation regions with 50% overlap. This rendered a spatial resolution of approximately 2 mm. The velocity statistics were checked to ensure that a significant spatial bias or "peak locking" was not taking place.

Results

The first experiments involved the use of the LSI from previous studies [7,9,11] to obtain base line information on H_2 operation. This LSI has a swirl number $S=0.54$ and is referred to here as LSI-R. Shown in Fig. 3 is the operating regime of LSI-R for $10<U_0<22$ m/s. Here, the LBO limit is nearly constant at $\phi\approx0.17$ and is consistent with the result reported previously for $U_0<10$ m/s [11]. This shows again that the LBO of LSI is relatively insensitive to the Reynolds number.

The LSI-R H_2 flames propagate closer to the exit than the hydrocarbon flames. In addition, there is an upper limit shown by the broken line at $\phi\approx0.32$. Above this boundary, the trailing edges of the flames curl back upstream and attach to the burner rim. This phenomenon indicates burning within the mixing layer formed between the reactants and the ambient air. It is a consequence of the high diffusivity and flammability of H_2.

Higher H_2 concentrations in these richer mixtures promote H_2 diffusion into the outer region of the mixing layer. The outer low velocity region becomes flammable, allowing the trailing edge of the flame to burn back upstream toward the rim. Flame attachment also occurs in CH_4/H_2 flames when the concentration of H_2 is high (>50) but does not occur in pure hydrocarbon flames. The slight upward trend of the flame attachment limit, as shown in Fig. 3, suggests that the phenomenon may not occur at high velocities.

When the flame attaches to the burner rim, it forms an envelope over the reactants. Consequently, the flow field and the flame/flow dynamics will be different from those of lifted flames. The close proximity of the H_2 flames in LSI-R to the exit also restricts laser access to the near field regions where information is needed to characterize the flame and flow properties. Therefore, LSI-R is not ideally suited for the current study and the swirl number needs to be lowered to relax the axial aerodynamic stretch rate so that the H_2 flames lift off farther from the exit.

Table 2 Perforated plates to vary the swirl number of LSI

	Hole diameter (mm)	S
LSI-R	3.05	0.54
LSI-UH1	3.18	0.53
LSI-UH2	3.45	0.50
LSI-UH3	3.66	0.47
LSI-UH4	3.86	0.41

To lower the swirl number, a series of perforated plates with decreasing blockage ratios was fitted to the center channel of the LSI swirler. These plates were made by enlarging the holes of the perforated plate for LSI-R. LSI-R uses a standard perforated plate with holes in a hexagonal arrangement. The hole diameter is 3.05 mm with a center-to-center spacing of 4.8 mm to give a blockage ratio of 40%. Table 2 shows the different plates used to produce a set of LSIs (designated as UH for uniform holes) with $0.53<S<0.41$.

Tests performed on the four LSI-UHs with H_2 air mixtures confirm that the flame liftoff distance increases with decreasing S as indicated by Eq. (2). Figure 4 compares their operating maps with that of LSI-R. The results show that a reduction of S from 0.54 to 0.50 has little effect on the LBO limits. Further reduction in S (<0.5) shifts LBO to higher ϕ and degrades the performance. The corresponding flame attachment limits show some changes but do not suggest a consistent upward trend with decreasing S. As indicated by the lengths of the vertical lines marking the extent of the effective operating regimes, decreasing S using these perforated plates tends to reduce the range of lifted H_2 flame operations.

In an effort to mitigate the H_2 flame attachment phenomenon, we applied a flame shaping method to elongate the H_2 flames so that their trailing edges are slightly swept back to reduce its propensity to ignite the mixing layer. This is accomplished by progressively increasing the size of holes on the center-channel perforated plate from small in the middle to large at the edge. These plates generate a slightly nonuniform axial velocity profile that is faster at the edges of the center channel, thus producing a slightly curved flame front at the center instead of a relatively flat flame when a UH plate is used.

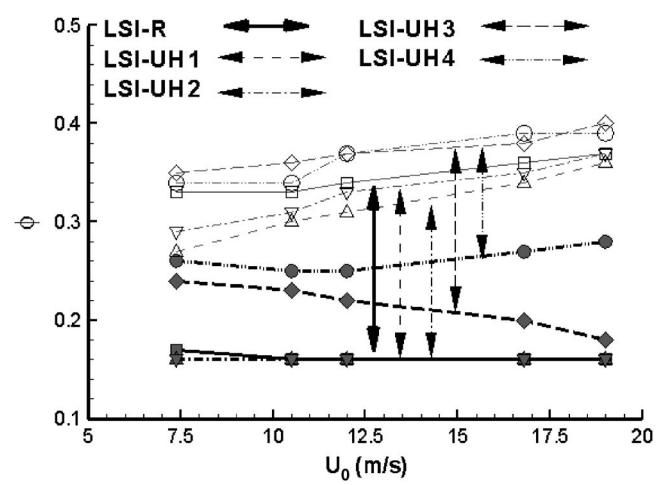

Fig. 4 Operating maps of LSIs with UH center plates

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Table 3 Hexagonal perforated plates with variable hole sizes to vary the swirl number of LSI

	Hole diameter (mm)			
	Center	Second and third	Fourth and fifth	S
LSI-VH1	3.05	3.26	3.45	0.51
LSI-VH2	3.26	3.45	3.57	0.50
LSI-VH3	3.05	3.45	3.66	0.48
LSI-VH4	3.05	3.7	3.8	0.43

Table 3 shows the four LSIs configured by varying hole (VH) diameter plates. These plates were made from the reference plate of LSI-R. The holes on these plates have three different diameters: one for the center hole, one for the second and third rings of holes from the center, and one for the fourth and fifth rings. The range of center-channel hole sizes for the LSI-VH is the same as that for LSI-UHs. Their swirl numbers $0.51 < S < 0.43$ are also similar.

Figure 5 shows the operating regimes of the LSI-VHs compared to that for LSI-R. It can be seen that the VH perforated

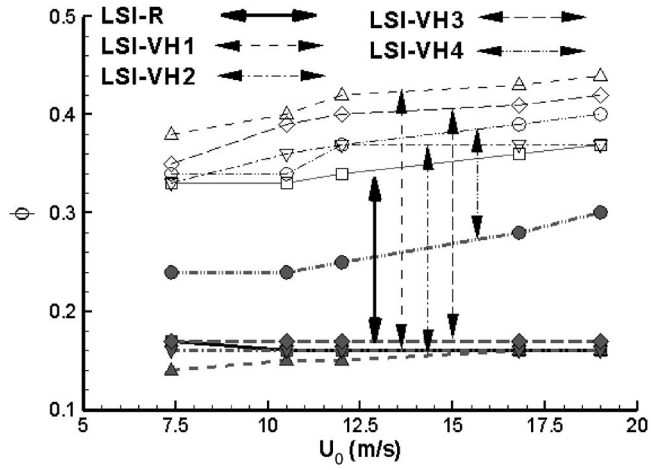

Fig. 5 Operating maps of LSIs with UH center plates

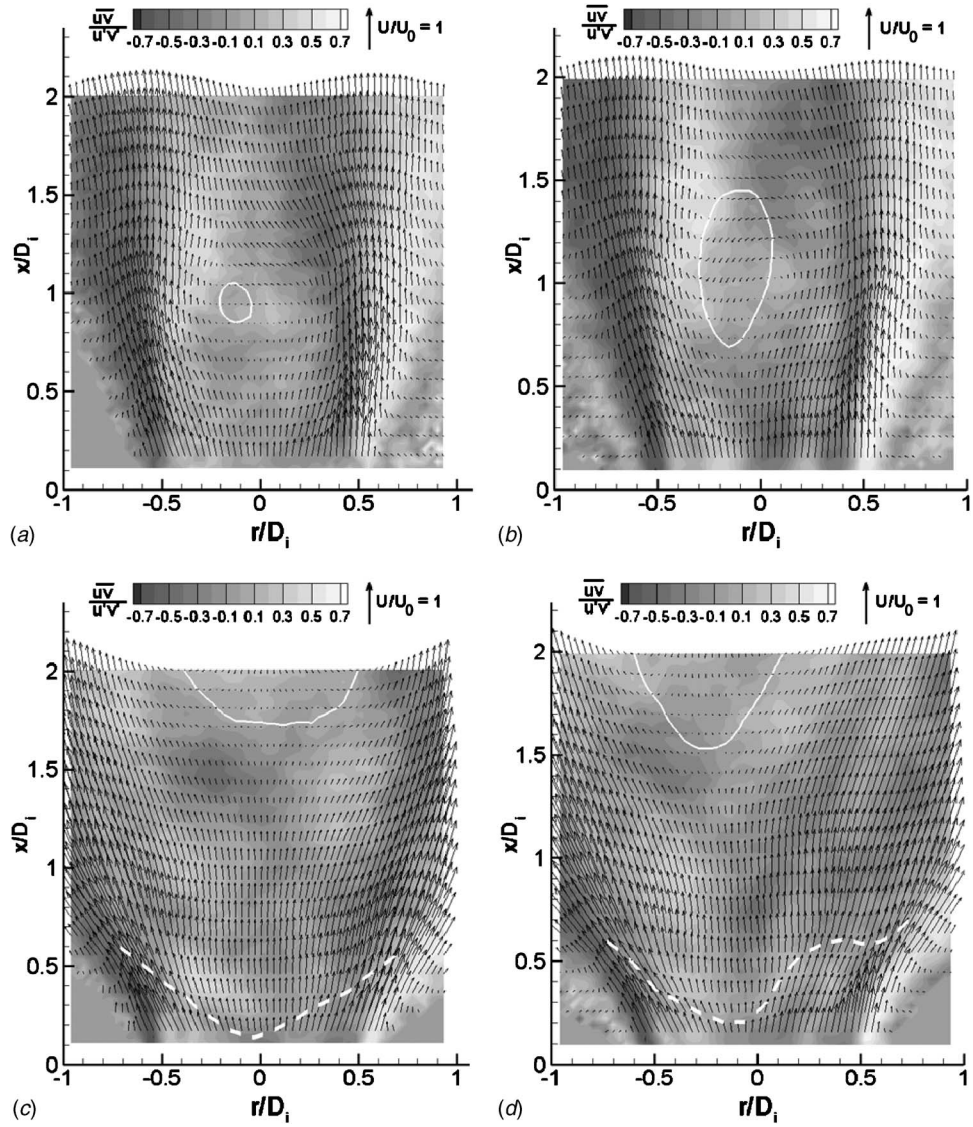

Fig. 6 Normalized 2D velocity vectors and normalized shear stresses measured in nonreacting flows of U_0=18 m/s in (*a*) LSI-R and (*b*) LSI-VH1 and in H_2/air flames of ϕ=0.35 and U_0 =18 m/s in (*c*) LSI-R and (*d*) LSI-VH. Dashed lines mark the leading edges of the flame brushes.

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plates improve the ranges of operation. Except for LSI-VH4, all the LBO limits are the same as LSI-R. More significantly, the flame attachment limits of some of the LSI-VHs are higher than that of the LSI-R. Therefore, flame shaping enables the study of lifted H_2 flame at higher stoichiometries. Consequently, the PIV measurements were made in the LSI-VHs and compared with those made in LSI-R.

The PIV measurements consist of 32 flames with $10 < U_0 < 20$ m/s using the mixtures of Table 1. Figure 6 compares the flow fields of LSI-R and LSI-VH1. From Fig. 5, LSI-VH1 with $S=0.51$ offers the largest range of operation among the LSI evaluated in this study and is considered the optimum configuration. The normalized 2D velocity vectors obtained in the nonreacting flows produced by LSI-R and LSIVH1 at $U_0=18$ m/s are shown, respectively, in Figs. 6(*a*) and 6(*b*). In the near field region of LSI-VH1 $(0.3 < x/D_i)$, a slight retardation in the flow velocities near the centerline is a direct consequence of the VH size plate. As outlined by the white line showing the zero velocity contour, both LSI-R and LSI-VH1 generate very weak recirculation zones in the far field. However, the one produced by LSI-VH1 is larger and elongated. Otherwise, there are no significant differences between the overall features of the nonreacting flow fields of the two LSIs.

The flow fields of H_2 flames at $\phi=0.35$ m/s and 18 m/s in LSI-R and LSI-VH1 are shown, respectively, in Figs. 6(*c*) and 6(*d*). At this stoichiometry, the leading edge of the flame brush of LSI-R is very close to the LSI exit at $x/D_i=0.2$. Therefore, the velocity data in Fig. 6(*c*) cannot show the details of the near field reactant region. However, in the far field, flow features such as combustion induced flow acceleration and an enlarged recirculation zone are similar to the features found in hydrocarbon flames. Hence, the overall changes in the flow fields induced by H_2 and hydrocarbon flames are not significantly different. The reacting flow field of LSI-VH1 is shown in Fig. 6(*d*). This flame has a larger liftoff $(x/D_i=0.3)$ but is asymmetric. The velocity vectors below the flame show noticeable nonuniform distributions because of the varying perforated plate hole sizes. Though the flow field is slightly off center, all other features of the velocity vectors, i.e., combustion heat release, flow discharge angle, and far field recirculation zone, are not significantly different from those shown in Fig. 6(*c*) for LSI-R.

According to Eq. (2), the flow fields of the LSI can be characterized by the parameters S_T, a_x, and x_0 deducible from the centerline velocity profiles. The correlation of the turbulent flame speed S_T is of particular importance because it quantifies the governing process of the LSI method. Turbulent flame speed data for H_2 are very rare and those reported in the literature show a nonlinear correlation with turbulence intensity u' commonly known as the bending effect. This so-called bending is also reported for S_T of hydrocarbon flames [15,16]. In contrast, S_T for hydrocarbon flames deduced from LSBs and LSI show a clear linear increasing trend with u' [11]. According to Eq. (2), the linear correlation is the significant flame property that enables the LSI to operate within a very wide range of flame and flow conditions. If the S_T correlation is nonlinear or bends, it implies that the flame drifts downstream with increasing U_0 and blows off.

Figure 7 compares the S_T data deduced from the 32 H_2 and H_2/N_2 flames with those reported previously for CH_4, C_3H_8, C_2H_4, and CH_4/H_2 [11]. The H_2 flame results show that their turbulent flame speeds are higher than those of the hydrocarbon flames. They are also more scattered, especially at the high u'/S_L conditions. The increase in scatter may be due to the uncertainties in the values of the laminar flame speed S_L for lean H_2 mixtures. Nevertheless, the H_2 turbulent flame speed data indicate a linear trend with increasing u'. The linear correlation constant K is 3.15 and is higher than the 2.1 value for hydrocarbon flames. The significant implication of the linear H_2 flame speed correlation is that the LSI concept should also be valid for hydrogen.

In addition to the linear flame speed correlation, the self-similarity of the near field region is the other important feature for the LSI operation. The two parameters in Eq. (2) that characterize the near field are the nondimensionalized axial aerodynamic stretch rate a_x and the virtual origin of the linear divergent region, x_0. The values of a_x deduced from the nonreacting and reacting flow fields are shown in Fig. 8 as functions of the flow Reynolds number, Re. The profiles in Fig. 8(*a*) show that the values of a_x

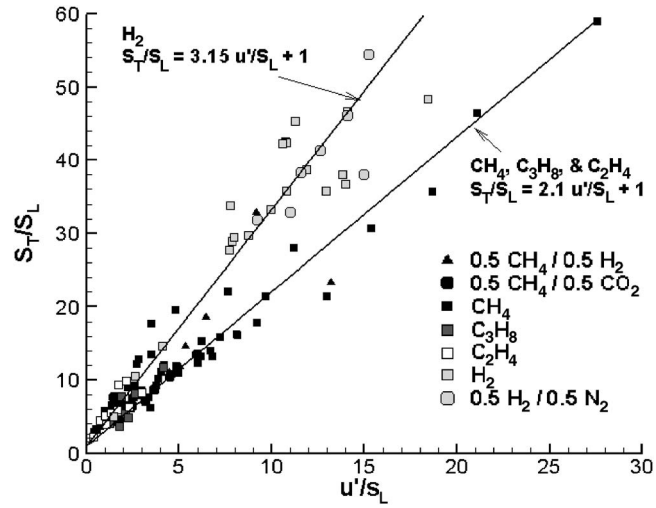

Fig. 7 Comparison of the turbulent flame speed S_T of H_2 and hydrocarbon flames

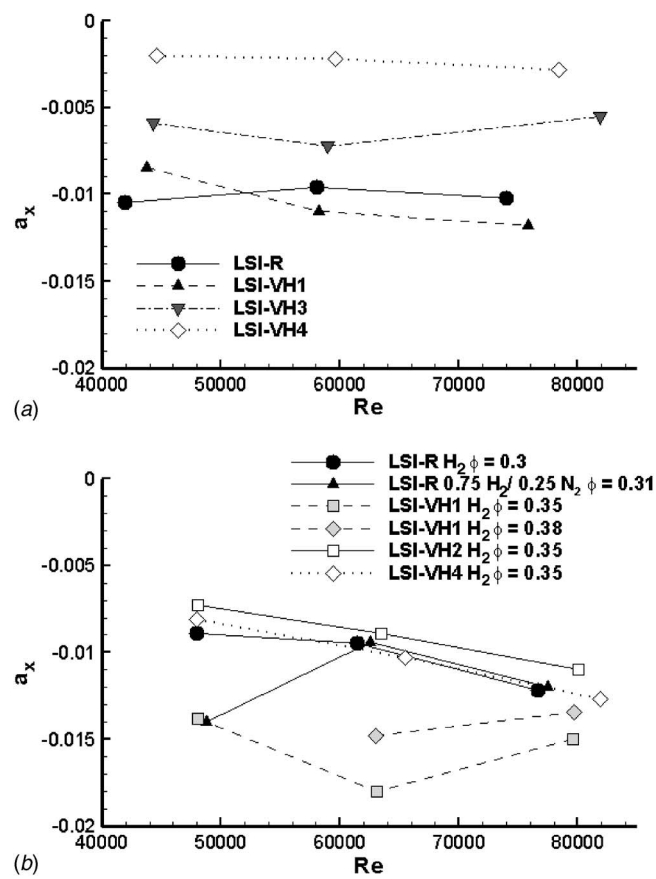

Fig. 8 Nondimensionalized axial aerodynamic stretch rate a_x measured in the nonreacting (*a*) and reacting (*b*) flow fields of the LSIs

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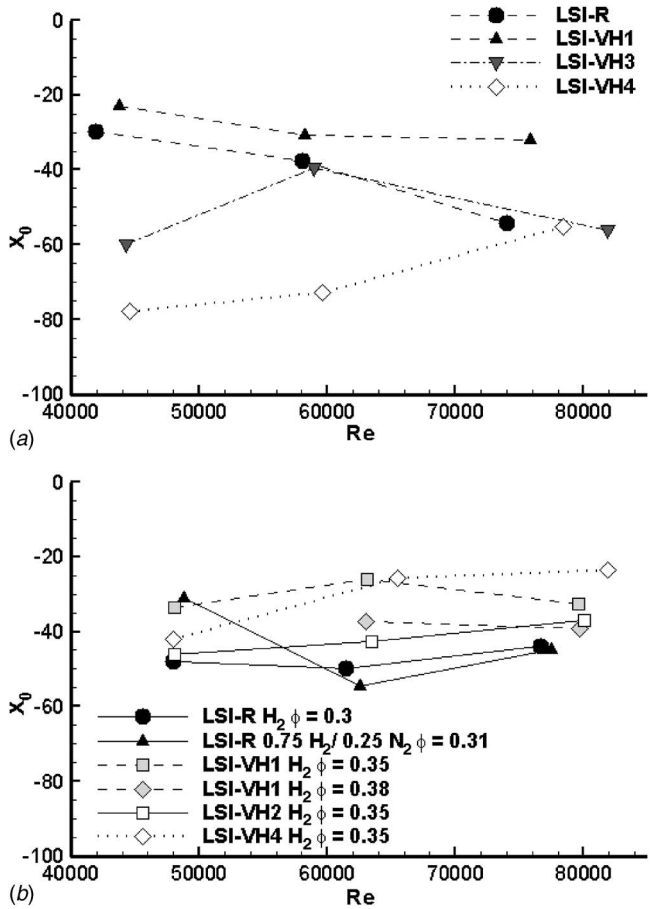

Fig. 9 Virtual origin x_0 deduced for the nonreacting (*a*) and reacting (*b*) flow fields of the LSIs

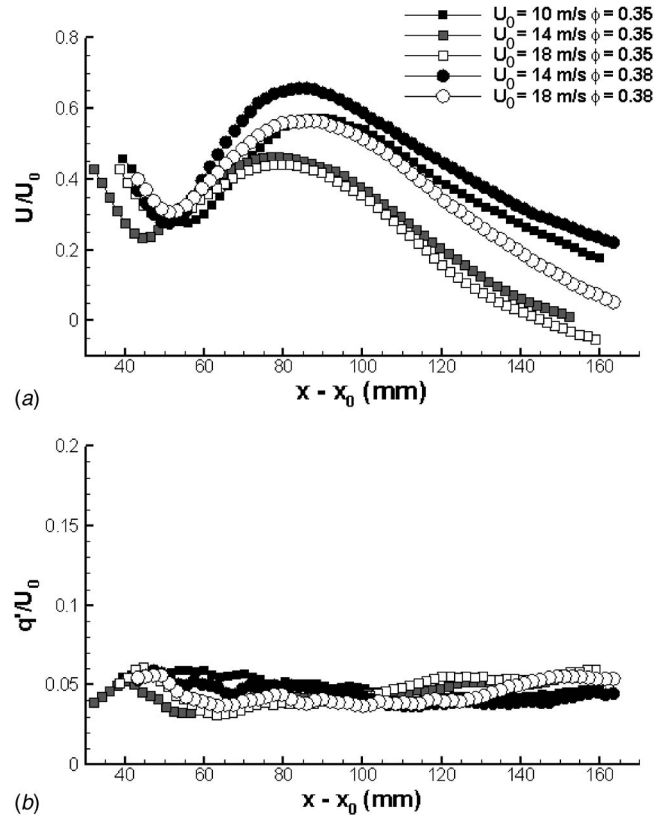

Fig. 10 Centerline velocity profiles of H_2 flames by LSI-VH1

for each LSI remain relatively constant within the range of Re investigated. This seems to indicate that the basic flow structures of these nonreacting flows are fully developed at Re > 40,000. There is also a dependency of the divergence rate (represented by the absolute value of a_x) on the increasing swirl number. The trend does not seem to be linear because a change from $S=0.43$ (LSI-VH4) to $S=0.51$ (LSI-VH1) reduces a_x from -0.003 1/mm to -0.01 1/mm. The corresponding a_x measured in H_2 flames are shown in Fig. 8(*b*). These values are generally lower than those of the nonreacting flows and indicate a systematic increase in the aerodynamic stretch rate induced by combustion heat release. This notion is supported by the fact that the more significant changes are found in flames with higher T_{ad} (e.g., $\phi \geq 0.35$).

The behaviors of the nonreacting and reacting a_x are generally consistent with those observed in the hydrocarbon flames. In addition, the data in Fig. 8(*b*) show a convergence to $a_x \approx -0.015$ 1/mm at high Re. This is the same level as those measured in CH_4 and other hydrocarbon flames [11]. The convergence trend seems to suggest that the flow structures in the near field regions of the reacting flows are still evolving but may become more similar at higher flow velocities.

The values of the virtual origins x_0 for the nonreacting and reacting flows are compared in Fig. 9. The nonreacting flow data of Fig. 9(*a*) show some scatter at Re < 50,000, with x_0 decreasing with S. At the higher Re range, the x_0 profiles tend to converge to about -60 mm. Together with the a_x data of Fig. 8(*a*), these results imply a systematic shift of the position of the divergent flow structure with increasing Re without a significant change in the divergence rate.

The results in Fig. 9(*b*) for the reacting flows show that the effects of H_2 flames on the LSI near fields are not very different from the effects of hydrocarbon flames. At Re > 70,000, the values of x_0 for all H_2 flames fall within a $-30 < x_0 < -50$ mm. These values are consistent with those reported in previous studies. Collectively, the present and previous data imply that with increasing U_0, the flow structures and the divergence rates of the near field produced by a LSI tend toward a self-similar structure that is independent of the flame property.

The self-similar behavior of the H_2 flame flow field is illustrated by the centerline velocity profiles (Fig. 10) and the transverse velocity profiles at $x=12$ mm (Fig. 11) of LSI-VH1. In Fig. 10(*a*), the features of the U/U_0 profiles are similar to those reported for hydrocarbon flames. All show linear velocity decays near the LSI exit whose slopes correspond to a_x. The positions where the profiles deviate from the linear decays correspond to the leading edges of the flame brushes. Further downstream, the increases in U/U_0 indicate combustion induced flow acceleration. A subsequent drop in U/U_0 in the far field is associated with the formation of the weak recirculation zone. In comparison, flow acceleration through the flame is not very pronounced on the velocity data obtained for a very lean flame ($\phi < 0.3$) burning a fuel mixture consisting of 50% CO_2 and 50% H_2 [11]. This shows that the effects of combustion heat release are manifested mainly in the far field.

The corresponding centerline profiles of the normalized turbulent kinetic energy q' show that they all show relatively flat distributions throughout the near field and the far field. The lack of increases in the far field near where the recirculations are formed is the only significant difference between the flow fields of the H_2 and the hydrocarbon flames.

The radial profiles at $x=12$ mm are shown in Fig. 11. This axial position is below the leading edge of the turbulent flame brushes, and the profiles are characteristic of the reactants entering the flame brushes. Within the central region, $-20 < r < 20$ mm, a

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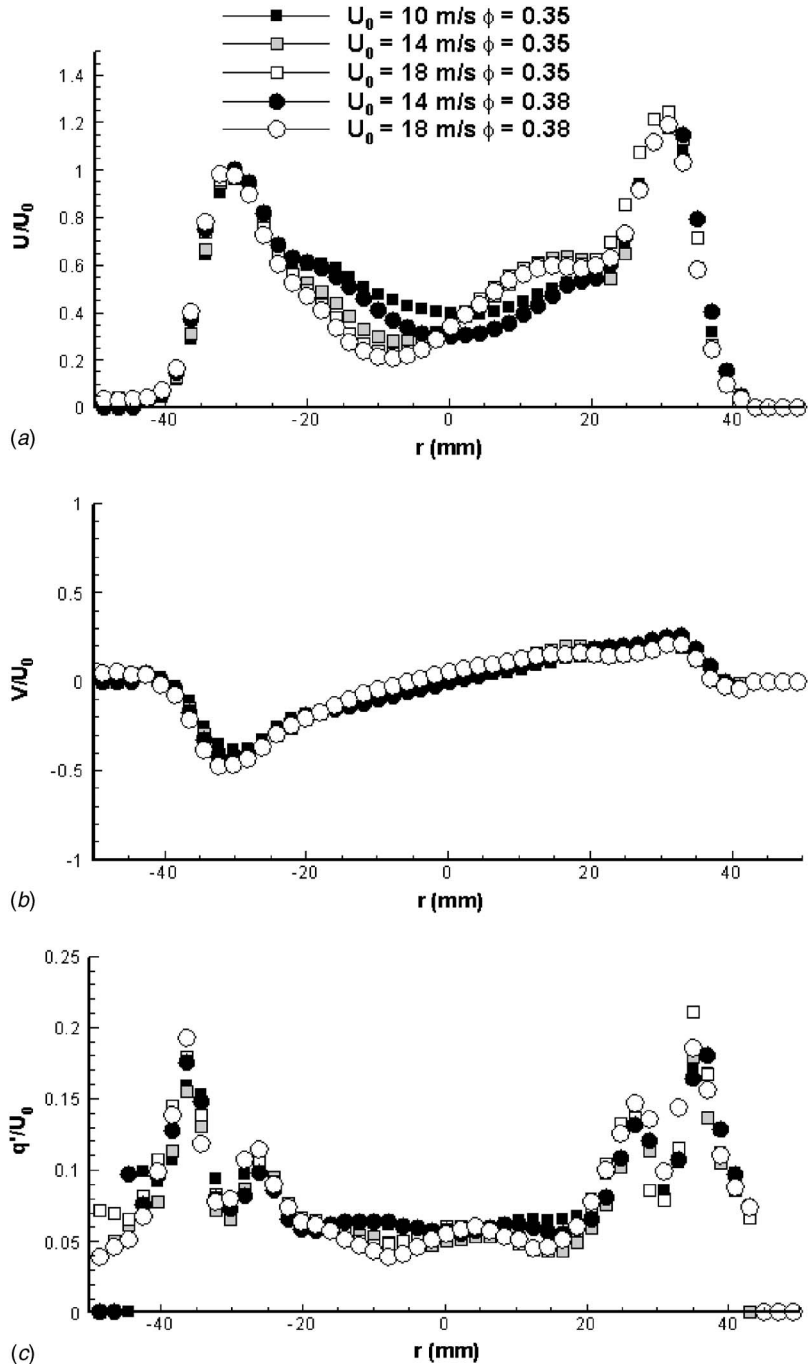

Fig. 11 Radial profiles at x=12 mm of H_2 flames by LSI-VH1

slight velocity deficit is clearly shown. This is a feature of the flow generated by these LSIs with VH plates. The slight asymmetries of the flow fields of three flames are also noticeable, but the differences in the U/U_0 levels within the central region are not any higher than those reported for hydrocarbon flames. In Fig. 11(b), all of the V/U_0 profiles collapse into a consistent distribution. The corresponding profiles of the turbulent kinetic energy q'/U_0 in Fig. 11(c) shows some scatter within the central region. However, all other features remain the same as those observed in hydrocarbon flames. The axial and radial velocity profiles of Figs. 10 and 11 show the self-similar nature of the near field regions of the H_2 flame flow fields.

It is very encouraging that the H_2 flames have near fields that are self-similar and turbulent flame speeds that correlate linearly with u'. These results show that the LSI concept will be amenable to burning of H_2 fuels. However, due to the complication caused by flame attachment, our investigation is limited to relatively lean H_2 mixtures of $\phi \leqslant 0.38$. A more extensive investigation at higher stoichiometries and bulk flow velocities is necessary to obtain additional data to support the development of the analytical model for guiding the design of a LSI prototype suitable for evaluation at high temperature and pressure with H_2.

Conclusions

Laboratory experiments have been conducted to obtain the basic information needed to guide the adaptation of the LSI to burn pure H_2 and N_2 diluted H_2 fuels that will be utilized in the gas

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turbines of the IGCC coal power plants. Starting with the LSI-R ($S=0.54$) developed for natural gas as a reference, the LBO limits for H_2 and H_2 diluted with N_2 were determined. The results show that LSI-R supports very lean flames ($\phi \approx 0.17$) and the blowoff limits are independent of velocity within the range investigated ($10 < U_0 < 20$ m/s). However, the LSI-R is less than optimum for the open H_2 flame experiments because of flame reattachment to the LSI-R rim at $\phi > 0.3$. The attached flame phenomenon is an impediment for the investigation of near field flow features of lifted flames. This problem was addressed by a flame shaping method that involved the use of center channel plates with VH sizes. Four LSI configurations with the VH plates of $0.41 < S < 0.51$ were evaluated. The best performance was provided by LSI-VH1 with $S=0.51$ that produced a lifted H_2 flame at $\phi=0.4$.

Extensive PIV measurements were made in 32 H_2 flames generated by LSI-R and four LSI-VHs. The experimental conditions consist of U_0 from 10 m/s to 20 m/s and ϕ from 0.3 to 0.38. The adiabatic flame temperatures of these H_2 flames are below 1400 K, and the NO_x formation is expected to be below the detectable limit. Consequently, emission measurements were not made in these flames.

The PIV results showed that the overall flow field features of the H_2 flames are not significantly different from those of the hydrocarbon flames. The turbulent flame speeds of the H_2 flames deduced from the centerline velocity profiles show a linear correlation with turbulence intensity u'. The value of the correlation constant of 3.1 is 50% higher than the value determined for hydrocarbon flames. A comparison of the near field features below the lifted H_2 flames was based on examining the changes in the normalized axial aerodynamic stretch rate a_x and the virtual origin x_0 with flow Reynolds numbers. The results show that the near field of the H_2 flames is self-similar. The values of a_x and x_0 for H_2 flames are similar to those found in hydrocarbon flames. Therefore, the basic LSI mechanism is not affected by the differences in the basic flame properties of hydrocarbons and H_2. The results of our laboratory studies show that the LSI concept is amenable to burning of H_2 fuels.

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Nomenclature

a_x = normalized axial aerodynamic stretch rate (l/mm)
D_i = injector diameter (mm)
L_i = swirler recess distance
$m=m_c/m_s$ = mass flux ratio
m_c = mass flux through the center channel
m_s = mass flux through the swirl annulus
q' = 2D turbulent kinetic energy $=1/2(u'^2+v'^2)^{1/2}$
Re = Reynolds number $2R_iU_0/\nu$
$R=R_c/R_i$ = ratio of the center-channel radius R_c to the injector radius R_i
r = radial distance
S = swirl number
S_L = laminar flame speed
S_T = turbulent flame speed
T_{ad} = adiabatic flame temperature
U_0 = bulk flow velocity
U = axial velocity
u' = axial rms velocity
v' = radial rms velocity
uv = shear stress
x = axial distance from the injector exit
x_f = leading edge position of the flame brush
x_0 = virtual origin of the divergent flow

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